PROJECT CASE STUDY

Weighted YOLO

Unified Multi-Generation Detection Architecture

Category	Status	Year
AI Research	Experimental research	2026

A proposed detector that studies YOLO evolution and combines weighted multi-scale fusion, modern CSP blocks, training-only gradient modules, and an end-to-end NMS-free head.

P2-P6	Weighted	NMS-free
feature scales	PAN-FPN fusion	target inference

Context and problem

YOLO generations introduce useful ideas independently, but a controlled study is needed to test which combinations improve the speed-accuracy trade-off and small-object detection.

Strategy and implementation

- Reviewed modules introduced across YOLOv4/v5/v8/v9/v10/YOLO11 and later designs.
- Designed a CSP-C3k2 backbone with SPPF, attention, and optional P2/P6 scales.
- Replaced equal feature merging with learnable weighted PAN-FPN fusion.
- Planned a one-to-many training head and one-to-one NMS-free inference path with ablations.

My role

Architecture research, module comparison, implementation planning, weighted-fusion experiments, logging diagnosis, and ablation design.

Technology stack

PyTorch | Ultralytics | YOLO | C3k2 | C2PSA | PAN-FPN | PGI

Results and evidence

- Produced a complete architecture and part-by-part technical explanation.
- Implemented learnable weighted neck fusion in matched experiments.
- Maintained research-safe claims: the unified model remains experimental until longer ablations establish superiority.

Key takeaways

- A novel architecture claim requires matched baselines and repeated ablations.
- Training-only modules should improve optimization without increasing inference cost.

Research note: This project is presented with conservative claims. Proposed architecture elements and prototype evidence are separated from results that still require broader comparison or validation.

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